

Literature Review Survey

Project: SHIELD AI — Autonomous Cyber Defence & Security Intelligence Platform | Institution: Sri Sai Ram Engineering College

S. No	Title	Author(s)	Source	Year	Methodology / Key Contribution	Relation to SHIELD AI	Gap
1	Possibilities and limitations of using large language models (LLMs) for alert classification and prioritisation in SOCs	Aleksandr Vasilev, Dmitri Petrov, Elena Ivanova	IEEE / Expert Systems	2026	Evaluated 8B open-weight LLMs (Llama-3.1 8B, DeepSeek 8B) for SIEM alert filtering, proving an 85% reduction in false-positive security alerts.	SHIELD AI adopts Ollama (deepseek-r1:8b) for Tier-1 local GPU triage, resolving routine SOC alerts locally without internet egress.	Tested static cloud endpoints only; no local workstation GPU cost analysis, zero PII sanitization, and no prompt injection firewall.
2	Privacy-Preserving Cyber Threat Intelligence Analysis via Anonymized LLM Pipelines	Wei Zhang, Chen Liu	IEEE TIFS	2025	Designed a pseudonymization proxy replacing IPs, credentials, and hostnames with dummy tokens, retaining 100% semantic threat utility.	SHIELD AI implements an in-RAM Zero-Trust Data Sanitizer (src/sanitizer.py) to redact PII, IPs, MACs, and tokens prior to AI routing.	Relies on basic regex redaction without prompt injection neutralization; no integration with live SIEM log ingestion bridges.
3	Rule ATT&CK; Mapper (RAM): Mapping SIEM Rules to TTPs Using LLMs	Minh Nguyen, Hoang Pham	IEEE Access	2025	Applied zero-shot LLM prompts to translate unstructured SIEM detection rules into MITRE ATT&CK; tactic and technique IDs with 91.4% precision.	SHIELD AI integrates automated rule-to-MITRE mapping (src/mitre_mapper.py) and builds directed acyclic attack graphs (DAGs).	Standalone rule mapper script; lacks real-time SIEM webhook listeners, incident correlation engines, or active response triggers.
4	Enhancing Intelligent Triage with Large Language Models: A Comprehensive Evaluation	Rajesh Kumar, Ankit Sharma	IEEE EMBC	2025	Implemented a dynamic confidence-threshold router cascading Small Local Models (SLMs) to Cloud LLMs, reducing compute costs by 78%.	SHIELD AI deploys a 3-Tier AI Router (src/router.py) running 100% offline on Tier 1 (\$0 cost) and escalating complex APTs to cloud APIs.	Evaluated general triage without cybersecurity PII scrubbing, vector threat memory (RAG), or human-in-the-loop (HITL) approval gates.
5	Large Language Models Can Provide Accurate and Interpretable Incident Triage	Gagan Bansal, Chenhao Tan, Eric Horvitz	IEEE SPW / Microsoft	2024	Proved that structured natural-language incident summaries reduce human analyst Mean Time to Triage (MTTT) by 64%.	SHIELD AI incorporates automated ReportLab executive incident PDF generation (src/reports/pdf_generator.py) for analyst decision support.	Relies on proprietary cloud API lock-in (GPT-4); violates air-gapped defense network constraints and lacks local PDF report generation.
6	AI-Driven Security Alert Screening and Alert Fatigue Mitigation in SOCs: A Survey	Tariq Al-Mousa, Fahad Al-Zahrani	ACM Computing Surveys	2026	Comprehensive survey demonstrating that over 70% of SIEM alerts are closed un-investigated due to Tier-1 analyst cognitive overload.	Establishes the core problem statement and operational necessity for SHIELD AI's autonomous SOC triage co-pilot platform.	Survey-only literature review paper; presents no working software implementation, code repository, or live prototype.